SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr.T.P.Anithaashri

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario.

(5 Marks)

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

(5 Marks)

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

(5 Marks)

Code of Conduct:

I K.Vijay Bhaskr Reddy(192425155) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Vijay Bhaskar Reddy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Answer:

The reward values increase gradually from **10 to 45** as the number of training episodes increases. This is a **positive learning trend** and indicates that the agent is improving its performance through repeated interaction with the environment.

At the beginning of training, the agent has limited knowledge about the environment. Therefore, it may select inappropriate actions and receive low rewards. As the number of episodes increases, the agent learns from its previous actions and adjusts its **policy**, which is the strategy used to select actions.

The gradual increase in reward means that the agent is increasingly selecting actions that produce better outcomes. The increase from 10 to 45 is particularly important because it shows a substantial improvement rather than random variation.

If the reward continues to increase and eventually becomes stable around a high value, it indicates that the policy is approaching **convergence**. This means that the agent has learned a consistent strategy and further training produces only small improvements.

Conclusion:

The increase in reward from **10 to 45 clearly demonstrates successful learning and policy improvement**. It indicates that the agent is making increasingly effective decisions and is moving toward a **stable and potentially near-optimal policy**.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Answer:

The **epsilon-greedy strategy** controls the balance between **exploration and exploitation**.

- **Exploration** means trying different actions to discover potentially better choices.
- **Exploitation** means selecting the action that currently has the highest estimated value.

When **ε = 0.1**, the agent explores only about 10% of the time and exploits the best-known action most of the time. Therefore, once the agent has learned useful actions, it can use them frequently and obtain a **high cumulative reward**.

When **ε = 0.3**, exploration increases. This allows the agent to discover new actions but also causes it to choose some actions that may produce lower rewards.

When **ε = 0.5**, the agent explores approximately half of the time. As a result, it spends much more time taking actions that may not be optimal. This explains why the cumulative reward is the **lowest** at ε = 0.5.

The observed relationship between epsilon and reward shows that excessive exploration reduces short-term performance in this particular environment. A smaller epsilon is more effective because the agent has already learned useful actions and should exploit them more frequently.

Conclusion:

ε = 0.1 provides the best performance because it gives the agent a better balance between exploration and exploitation. The highest cumulative reward at ε = 0.1 confirms that **greater exploitation is more beneficial in the given experiment**.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

Answer:

Both **Q-Learning** and **SARSA** are value-based Reinforcement Learning algorithms, but they use different learning approaches.

Q-Learning is an off-policy algorithm. It learns the value of the best possible next action and aims to estimate the optimal action-value function.

SARSA is an on-policy algorithm. It learns using the actual action selected by the current policy. Therefore, its learning is directly influenced by the agent's behavior, including exploration.

In the given data, Q-Learning produces **slightly higher average rewards at 100, 200, and 300 episodes**. The important point is that this advantage is consistent across all three training stages. This indicates that Q-Learning is learning a better-performing policy in the given environment.

Since Q-Learning reaches and maintains higher rewards throughout training, it appears to achieve effective performance more quickly. However, true convergence speed is ideally established by observing when the reward becomes stable. Based on the provided observations, Q-Learning **appears to converge faster**.

Conclusion:

Q-Learning performs better because it consistently obtains **higher average rewards than SARSA**. The observed trend suggests that **Q-Learning reaches a high-performing policy faster**, making it the better-performing algorithm in this experiment.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

Answer:

The **discount factor (γ)** determines how much importance the agent gives to future rewards compared with immediate rewards.

When **γ = 0.2**, the agent strongly prefers immediate rewards and gives relatively little importance to future outcomes.

When **γ = 0.5**, the agent gives moderate importance to future rewards and considers both short-term and long-term consequences.

When **γ = 0.9**, the agent strongly considers future rewards. Therefore, it can learn that an action with a smaller immediate reward may be useful if it leads to a larger reward later.

The data shows that the **average reward increases as γ increases**. This indicates that long-term planning is important in this environment. The agent performs better when it considers the consequences of its actions over a longer period.

Therefore, the highest discount factor among the tested values provides the best performance.

Conclusion:

γ = 0.9 is the most suitable value because it produces the highest average reward. The results indicate that giving greater importance to **future rewards improves the agent's overall learning performance and long-term reward optimization**.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Answer:

The success rate increases from **45% at 50 episodes to 92% at 200 episodes**. This is a significant improvement of **47 percentage points**, showing that the agent has learned substantially from additional training.

At 50 episodes, the agent succeeds in less than half of the attempts. This indicates that the agent is still exploring the environment and has not yet learned an effective strategy.

At 200 episodes, the success rate reaches 92%. This means that the agent successfully completes the task in most attempts. The large increase demonstrates that the learned policy has become much more effective.

However, a 92% success rate does not by itself prove that the policy is completely optimal. An optimal policy would require the agent to consistently achieve the best possible performance. More training episodes and a stable success rate close to the theoretical maximum would provide stronger evidence of optimality.

Conclusion:

The increase from **45% to 92% clearly demonstrates strong policy improvement and learning progress**. The agent has developed a highly effective policy and is **approaching an optimal policy**, although complete optimality cannot be confirmed from the given data alone.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

Answer:

The **learning rate** determines the size of the update made to the agent's learned values during training.

With **learning rate = 0.01**, the updates are very small. This usually results in stable but **slow learning**, because the agent changes its estimates only gradually.

With **learning rate = 0.05**, the updates are moderate. The agent can learn at a reasonable speed without making excessively large changes to its value estimates. The given result shows that this value provides **higher stability**.

With **learning rate = 0.1**, the updates are much larger. Although this can speed up learning initially, it can also cause the value estimates to change too rapidly, producing unstable or oscillating learning.

Therefore, 0.05 provides the most appropriate balance between learning speed and stability.

Conclusion:

Learning rate = 0.05 is the best choice because it provides the best trade-off between **convergence speed and stability**. The observed higher stability indicates that the agent can learn effectively without excessive fluctuations.

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario.

Answer:

The three strategies differ in how they balance exploration and exploitation.

ε-greedy chooses the best-known action most of the time and randomly explores other actions with a probability of ε. It therefore gives strong preference to actions that have already produced good rewards.

Softmax exploration selects actions probabilistically according to their estimated values. Better actions have a higher probability of being selected, but lower-valued actions may still be chosen.

Random exploration selects actions randomly without strongly considering the learned action values. This can cause the agent to spend many steps taking actions that produce low rewards.

The experiment shows that the **cumulative reward is highest with ε-greedy**. This indicates that the environment benefits from strong exploitation of the best-known actions while still allowing some exploration.

The high cumulative reward also means that ε-greedy is using the learned Q-values more effectively than the other strategies under the given conditions.

Conclusion:

ε-greedy achieves the best performance because it provides an effective balance between exploration and exploitation. The highest cumulative reward demonstrates that this strategy is the most suitable among the three for the given Q-learning experiment.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

Answer:

The number of steps required to reach the goal is a measure of the agent's **efficiency**. Fewer steps generally indicate that the agent has learned a shorter and better path.

Initially, the agent takes **50 steps** to reach the goal. This indicates that it has limited knowledge of the maze and is spending many steps exploring unnecessary paths.

After **300 episodes**, the number of steps decreases to **15**. This is a major improvement and shows that the agent has learned a much more efficient route.

The reduction from 50 to 15 steps means that the policy has improved significantly through training. The agent is now making better decisions and avoiding unnecessary movements.

However, this does not automatically prove convergence to an optimal policy. To confirm optimality, the agent should consistently take the shortest possible path over many additional episodes.

Conclusion:

The reduction from **50 steps to 15 steps strongly indicates policy improvement and better learning efficiency**. It suggests that the agent is approaching an optimal policy, and stable performance at 15 steps would provide stronger evidence of convergence.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

Answer:

The **discount factor (γ)** controls the relative importance of immediate and future rewards.

With **$\gamma = 0.4$**, the agent focuses more on immediate rewards and gives less importance to rewards that occur later.

With **$\gamma = 0.7$**, the agent considers future rewards more strongly, providing a balance between immediate and long-term outcomes.

With **$\gamma = 0.95$**, the agent gives very high importance to future rewards. This allows it to choose actions that may provide a smaller immediate reward but lead to better long-term results.

The given data shows that the **highest long-term reward is obtained at $\gamma = 0.95$**. This means that the environment rewards long-term planning, and SARSA benefits from considering future consequences.

The result demonstrates that a higher discount factor is more appropriate when successful decisions depend on what happens over several future steps.

Conclusion:

$\gamma = 0.95$ is the most effective value because it produces the highest long-term reward. It allows the SARSA agent to balance immediate actions with future consequences and achieve better overall performance.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

Answer:

A **stochastic environment** contains randomness, meaning that the same action does not always produce exactly the same result. Therefore, fluctuations in reward are expected, especially during the early stages of training.

During the initial episodes, the agent has limited knowledge and performs considerable exploration. It tries different actions and receives different rewards, causing the reward curve to fluctuate.

After approximately **500 episodes**, the rewards become more stable. This indicates that the agent has learned which actions generally provide better outcomes and has developed a more consistent policy.

The stabilization is important because it suggests that the agent is no longer making large changes to its behavior after every episode. In other words, its policy is becoming more reliable.

In a stochastic environment, the reward may still show small variations because of randomness. However, a stable overall trend demonstrates improved **robustness, consistency, and convergence**.

Conclusion:

The early reward fluctuations indicate exploration and learning, while the stabilization after **500 episodes indicates significant policy improvement**. It shows that the agent has learned a reliable and robust strategy and is **approaching convergence despite environmental randomness**.